SANDEEP KUMAR KUMAWAT

M.Tech Researcher | Bioscience and Bioengineering | IIT Jodhpur

m25bbe007@iitj.ac.in | +91-8562013985

Sandeep kumar kumawat | LinkedIn | Website: sandeepkk.me

RESEARCH PROFILE

M.Tech researcher at IIT Jodhpur working on single-cell approaches to study cell-type-specific transcriptional responses associated with gene–environment interactions. Research experience includes single-cell analysis and an integrated analysis of three public gene-expression cohorts comprising 82 samples from idiopathic pulmonary fibrosis studies. Interested in computational methods for single-cell genomics, transcriptomic data integration and disease biology.

EDUCATION

M.Tech. in Bioscience and Bioengineering — **2025–2027 (Expected)**
Indian Institute of Technology Jodhpur, India | CGPA: 7.38 (current)

B.Tech. in Biotechnology — **Graduated 2025**
ASPEE SHAKILAM Biotechnology Institute, Navsari Agricultural University | CGPA: 8.02

Senior Secondary (Class XII) — **2016**
Board of Secondary Education, Rajasthan | 79.6%

Secondary (Class X) — **2014**
Board of Secondary Education, Rajasthan | 79.0%

RESEARCH EXPERIENCE

Indian Institute of Technology Jodhpur — **2025–Present**
M.Tech Thesis: Single-Cell-Based Approaches for G×E Analysis — *Jodhpur, India*

- Investigating cell-type-specific transcriptional patterns associated with gene–environment (G×E) interactions using single-cell transcriptomic data.
- Applying single-cell data preprocessing, quality control, clustering, dimensionality reduction, and cell-type annotation to support downstream analysis.
- Developing computational workflows to analyse high-dimensional single-cell data and identify biologically interpretable patterns relevant to G×E research.

Indian Institute of Technology Jodhpur — **May–July 2026**
Summer Thesis Research: Computational Analysis of Lung Fibrosis — *Jodhpur, India*

- Integrated three public GEO gene-expression cohorts comprising 82 samples across control, early-stage, and advanced idiopathic pulmonary fibrosis (IPF), applying ComBat for batch correction.
- Performed differential-expression analysis using limma across disease-stage comparisons and derived an 11-gene common signature.
- Characterised the gene signature using STRING protein–protein interaction analysis and g:Profiler pathway enrichment, identifying extracellular-matrix remodelling as a prominent biological theme.

MicroCare Laboratory — **Mar–May 2025**
Laboratory Intern — *Surat, India*

- Performed microbiological and pathological laboratory procedures on 500+ clinical samples, including culture/media preparation and handling of blood, urine, and sputum specimens.
- Conducted routine pathological procedures and microscopy, including simple and compound microscopy.
- Performed antimicrobial activity and antibiotic susceptibility testing across 20 antibiotics to evaluate microbial resistance patterns.

Navsari Agricultural University (NAU) — **Jul–Dec 2024**
B.Tech Thesis: Antibacterial Activity of Plant Extracts — *Surat, India*

- Prepared extracts of *Murraya koenigii* and *Trigonella foenum-graecum* using methanol, acetone, ethyl acetate, and petroleum ether, and quantified extractive yield.
- Evaluated antibacterial activity using agar well-diffusion assays against *B. subtilis*, *S. aureus*, *E. coli*, *S. typhi*, and *Serratia* across multiple concentrations.

TECHNICAL SKILLS

Programming & Computing: Python, R, Git, Linux/Ubuntu, Jupyter Notebook, RStudio

Single-Cell & Transcriptomics: scRNA-seq, Seurat, quality control, clustering, dimensionality reduction, cell-type annotation

Bulk Gene-Expression Analysis: GEO datasets, limma differential-expression analysis, ComBat batch correction, GSVA, CIBERSORTx

Statistics & Visualization: Correlation analysis, Wilcoxon rank-sum test, Kruskal–Wallis test, pandas, NumPy, SciPy, Matplotlib, ggplot2

Pathway & Network Analysis: g:Profiler, STRING, protein–protein interaction (PPI) analysis, Cytoscape

KEY COURSEWORK

Bioinformatics and Computational Biology; Genomics and Transcriptomics; Molecular and Cell Biology; Statistics and Data Analysis; Imaging Cell Biology

PUBLICATION

Vaghasiya, H. Y., & Kumawat, S. (2026). Solvent extraction-based screening of *Murraya koenigii* for antimicrobial efficacy against bacterial pathogens. In E. Siva Rami Reddy (Chief Ed.), *Advances in AYUSH* (Vol. 17, Chapter 9, pp. 119–128). AkiNik Publications. https://doi.org/10.22271/ed.book.3616

RESEARCH INTERESTS

Single-cell genomics | Computational biology | Gene–environment (G×E) interactions | Transcriptomic data integration | Multi-omics | Disease biology